

Yash Patel

Data Scientist

(224)-539-6564 | yasharvindpatel2000@gmail.com | <https://www.linkedin.com/in/yashpatel0301/> | <https://yashpatel301.github.io/>

SUMMARY

Data Scientist with expertise in machine learning, NLP, reinforcement learning, and AI-driven product development. Experienced in deploying scalable models, optimizing algorithms, and collaborating cross-functionally to deliver data-driven business insights.

EXPERIENCE

Orbe360, California, United States - *Data Scientist*

May' 25 - Present

- Developed data pipelines from mined sources and financial document scraping to populate Postgres databases and fine-tuned LLMs to build an in-house model. Engineered agentic chatbot workflows integrated into the ORBE platform.

DePaul University, Chicago, United States - *Research Assistant*

Apr' 24 - Dec' 24

- Refined resource allocation methods via Reinforcement learning model focused on optimizing computation tasks within distributed networks; directly improved user experience for thousands of concurrent connections relying on accurate processing times.

IQM Corporation, New York, United States - *Data Intern - Summer*

Jun' 24 - Aug' 24

- Conducted R&D on AWS QuickSight to build and deploy an extensive query resolver chatbot using LangChain, achieving 88+ BLEU accuracy in SQL generation and database retrieval for non-tech teams.

IQM Corporation, Ahmedabad, India - *Data Scientist*

Jan' 22 - Aug' 23

- Analysed large datasets using cosine similarity to assign 100,000-150,000 URL views to potential viewer IDs, enhancing data accuracy and user insights.
- Developed APIs for keyword extraction and implemented a BERT NLP model to predict categories for URLs, achieving efficient labelling of approximately 6,500 URLs every 12 hours.
- Partnered with the Data Engineering team, leading efforts from the Data Science side to develop a strategy that balanced ad recommendations by integrating insights from historical data with real-time incoming data streams.

Etasha Society, Delhi, India - *Data Analyst Volunteer*

Jan' 21 - Mar' 21

- Performed data cleaning, transformation, and reporting on program data, improving reliability by 30%.

SKILLS

- Programming:** Python, R, SQL, Java, C, C++
- ML/AI Frameworks:** Scikit-learn, TensorFlow, PyTorch, Hugging Face, LangChain, LangGraph
- Databases:** PostgreSQL, MySQL, Oracle
- Data & MLOps:** ETL, Feature Engineering, Model Deployment, Experiment Tracking, CI/CD, Docker, AWS SageMaker
- Analytics & BI:** Excel (PivotTables, VLOOKUP, Power Query), Power BI, Tableau, AWS QuickSight, Snowflake
- Techniques:** Machine Learning, Deep Learning, NLP, Generative AI, Reinforcement Learning, Computer Vision, A/B Testing, Predictive Modeling

PROJECTS

QA-Youtube-Videos

- Developed a Streamlit application leveraging GPT-3.5 with RAG architecture, LangChain tool calling, and FAISS-based semantic search to analyze YouTube video transcripts with support for multilingual queries and transcript translation.

Molecular Communication

- Designed and evaluated ANN and LSTM models for binary code prediction in a fluid medium, achieving 99.27% accuracy with LSTM. LSTM outperformed ANN by 0.3% in accuracy.

CERTIFICATIONS

- Machine Learning - Stanford University
- How Google does Machine Learning – Coursera
- Chatbot Workshop - The MAD
- Google Cloud Big Data and Machine Learning Fundamentals - Coursera

EDUCATION

DePaul University, Chicago, United States

Masters of Science in Computer Science

Sept' 23 - Mar' 25

Coursework: Programmatic Machine Learning, Artificial Intelligence, Data Mining, Database Systems

Ahmedabad University, Gujarat, India

Bachelor's in Information and Communication Technology

Aug' 18 - May' 22

Coursework: Machine Learning, Probabilistic Modelling, NLP, Computer Vision, Industrial Statistics